Digital Image Processing

COLOUR IMAGE PROCESSING

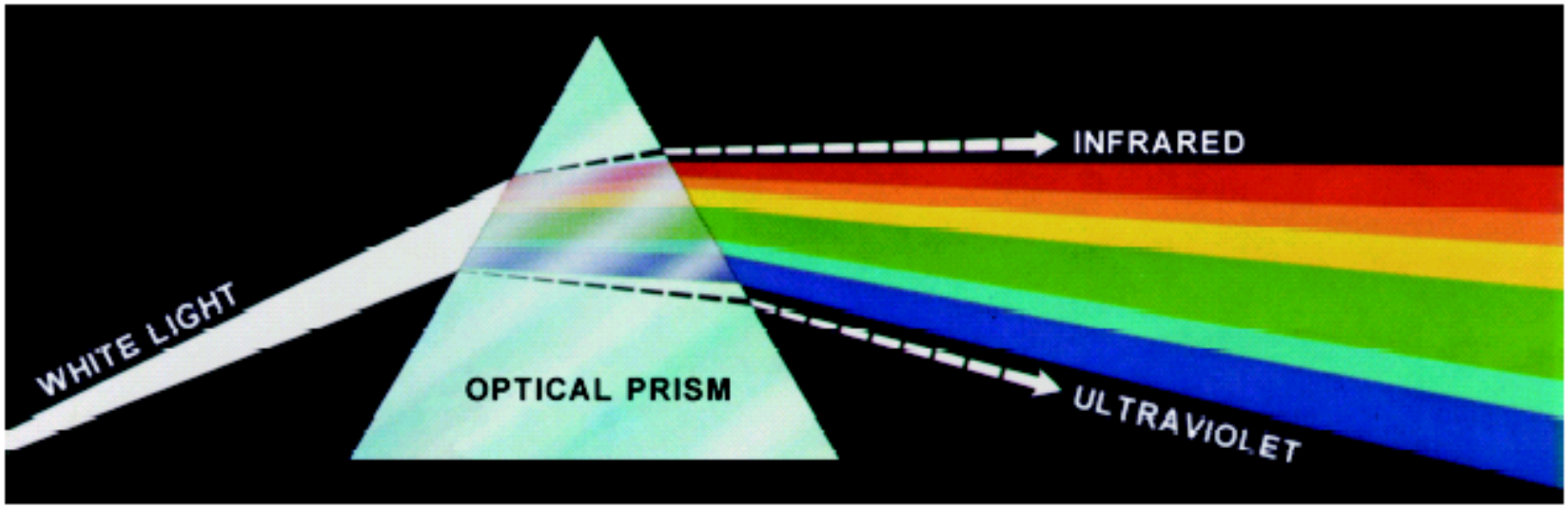
Introduction

Today we'll look at colour image processing, covering:

- Colour fundamentals
- Colour models

Colour Fundamentals

In 1666 Sir Isaac Newton discovered that when a beam of sunlight passes through a glass prism, the emerging beam is split into a spectrum of colours

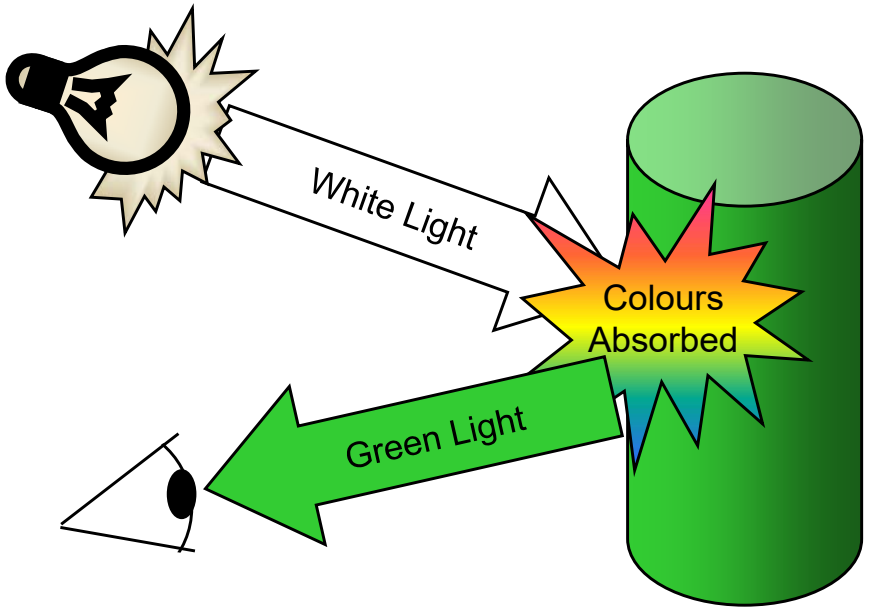


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Colour Fundamentals (cont…)

The colours that humans and most animals perceive in an object are determined by the nature of the light reflected from the object

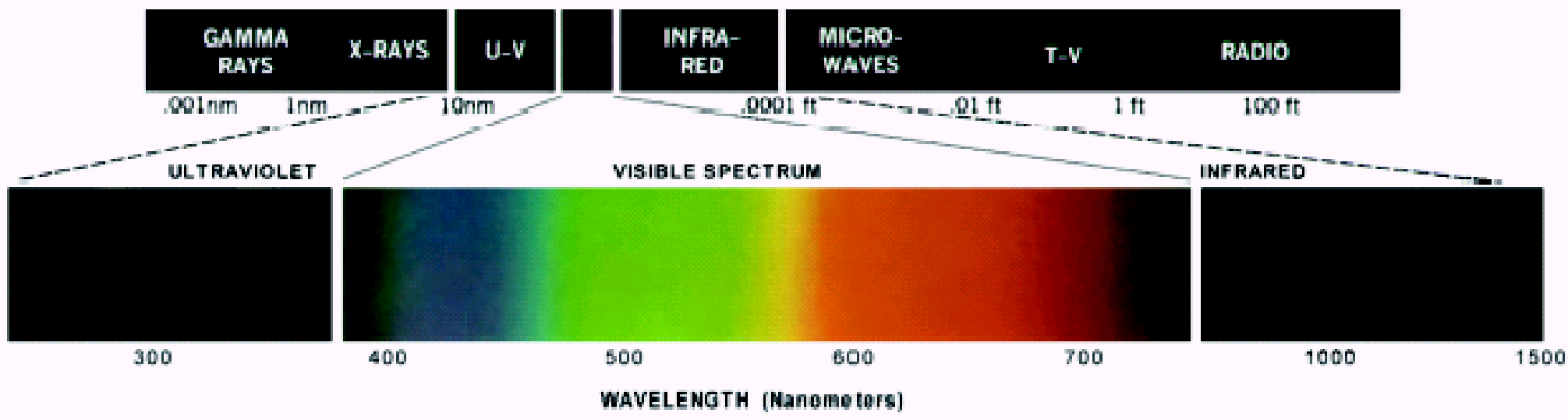
For example, green objects reflect light with wave lengths primarily in the range of 500 – 570 nm while absorbing most of the energy at other wavelengths



Colour Fundamentals (cont…)

Chromatic light spans the electromagnetic spectrum from approximately 400 to 700 nm

As we mentioned before human colour vision is achieved through 6 to 7 million cones in each eye



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

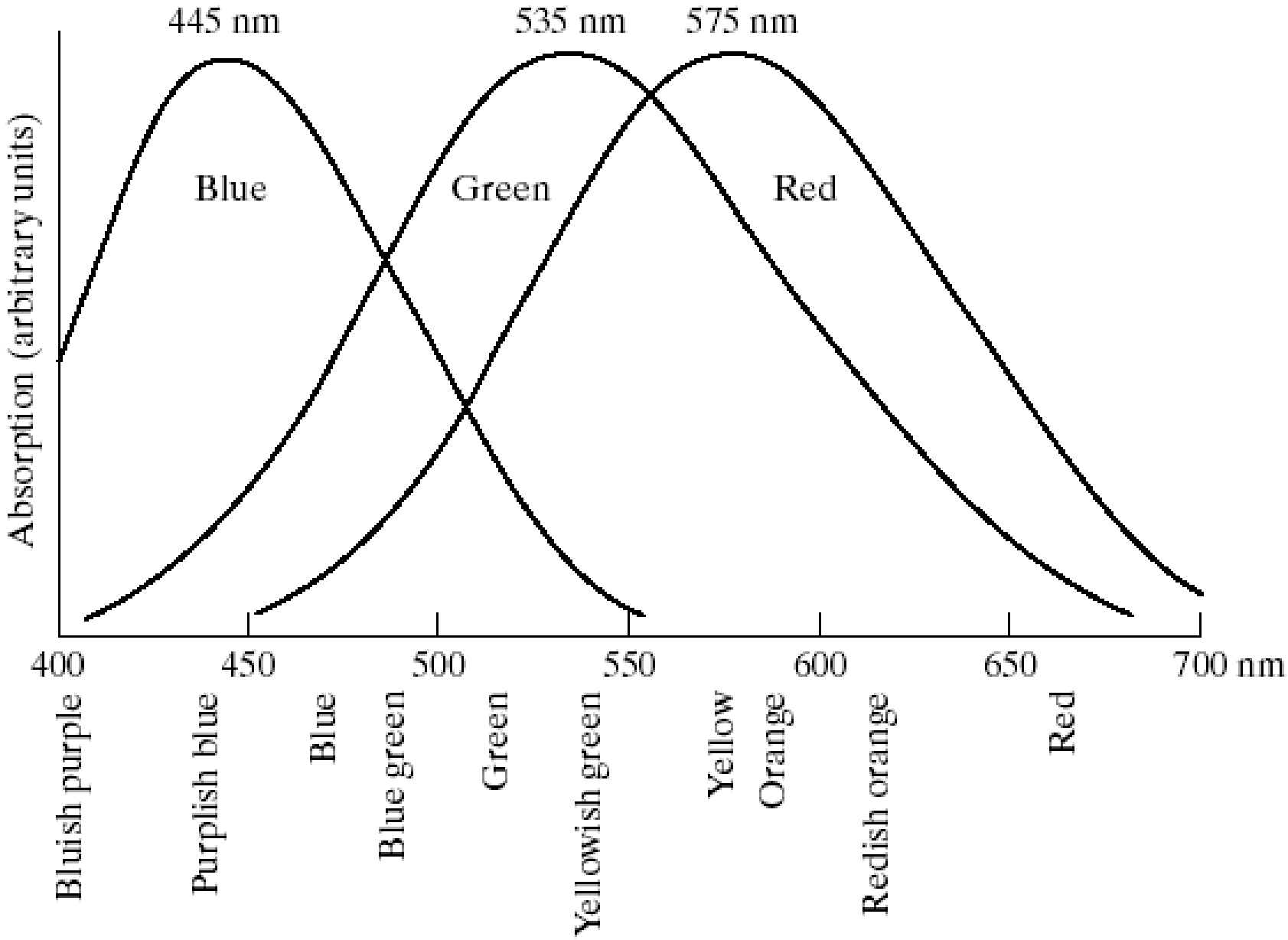
Colour Fundamentals (cont…)

Approximately 65% of these cones are sensitive to red light, 33% to green light and 2% to blue light

Absorption curves for the different cones have been determined experimentally

Strangely these do not match the CIE standards for red (700nm), green (546.1nm) and blue (435.8nm) light as the standards were developed before the experiments!

Colour Fundamentals (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Colour Fundamentals (cont…)

3 basic qualities are used to describe the quality of a chromatic light source:

- **Radiance:** the total amount of energy that flows from the light source (measured in watts)
- **Luminance:** the amount of energy an observer *perceives* from the light source (measured in lumens)
  - Note we can have high radiance, but low luminance
- **Brightness:** a subjective (practically unmeasurable) notion that embodies the intensity of light

We'll return to these later on

CIE Chromacity Diagram

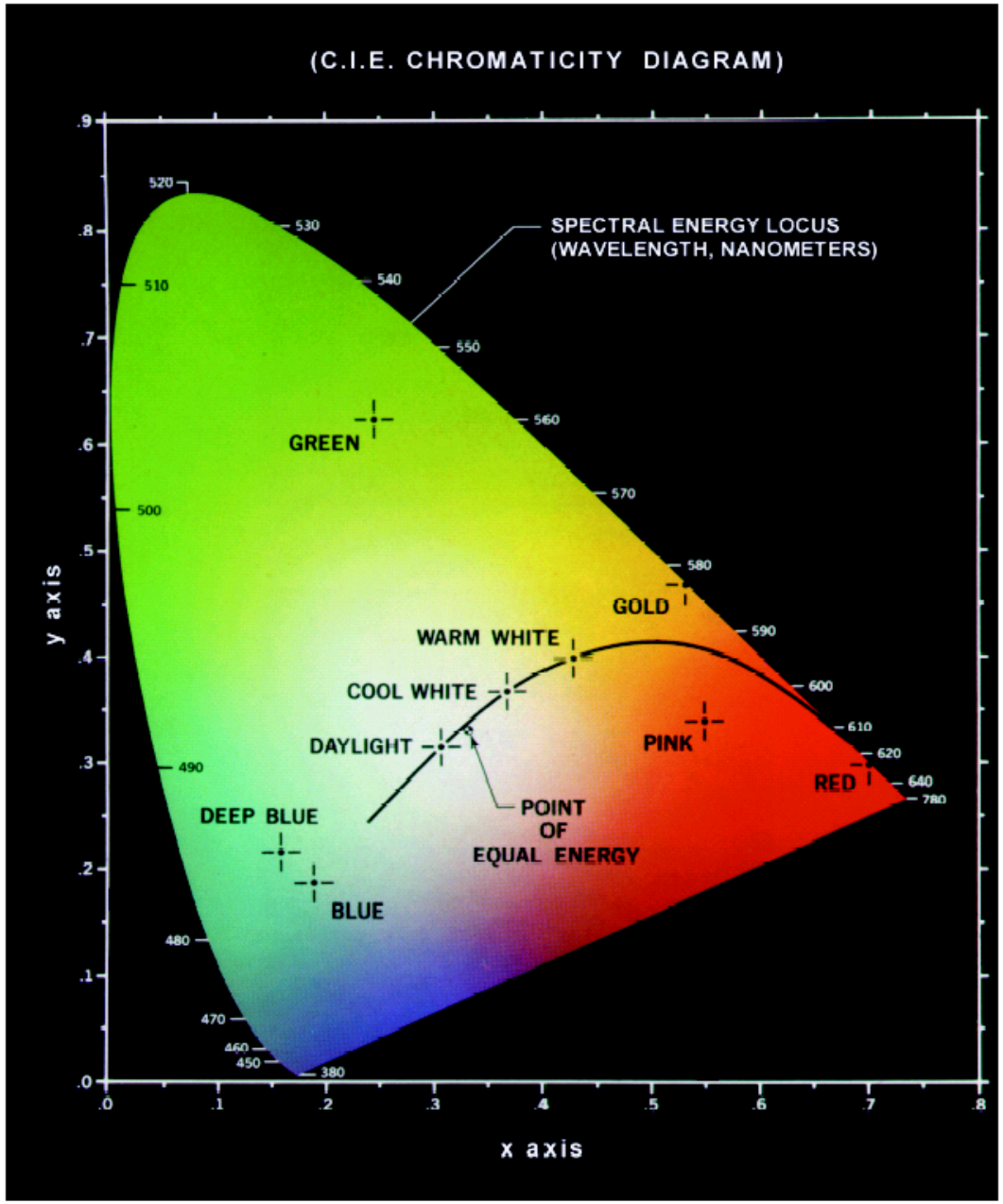
Specifying colours systematically can be achieved using the CIE **chromacity diagram**

On this diagram the x-axis represents the proportion of red and the y-axis represents the proportion of red used

The proportion of blue used in a colour is calculated as:

$$z = 1 - (x + y)$$

CIE Chromacity Diagram (cont…)



Green: 62% green, 25% red and 13% blue

Red: 32% green, 67% red and 1% blue

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

CIE Chromacity Diagram (cont…)

- Any colour located on the boundary of the chromacity chart is fully saturated
- The point of equal energy has equal amounts of each colour and is the CIE standard for pure white
- Any straight line joining two points in the diagram defines all of the different colours that can be obtained by combining these two colours additively
- This can be easily extended to three points

Colour Models

From the previous discussion it should be obvious that there are different ways to model colour

We will consider two very popular models used in colour image processing:

- RGB (**R**ed **G**reen **B**lue)
- HIS (**H**ue **S**aturation **I**ntensity)

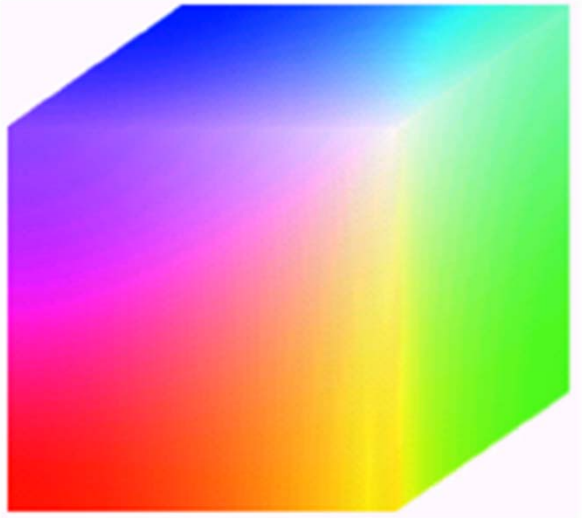
RGB

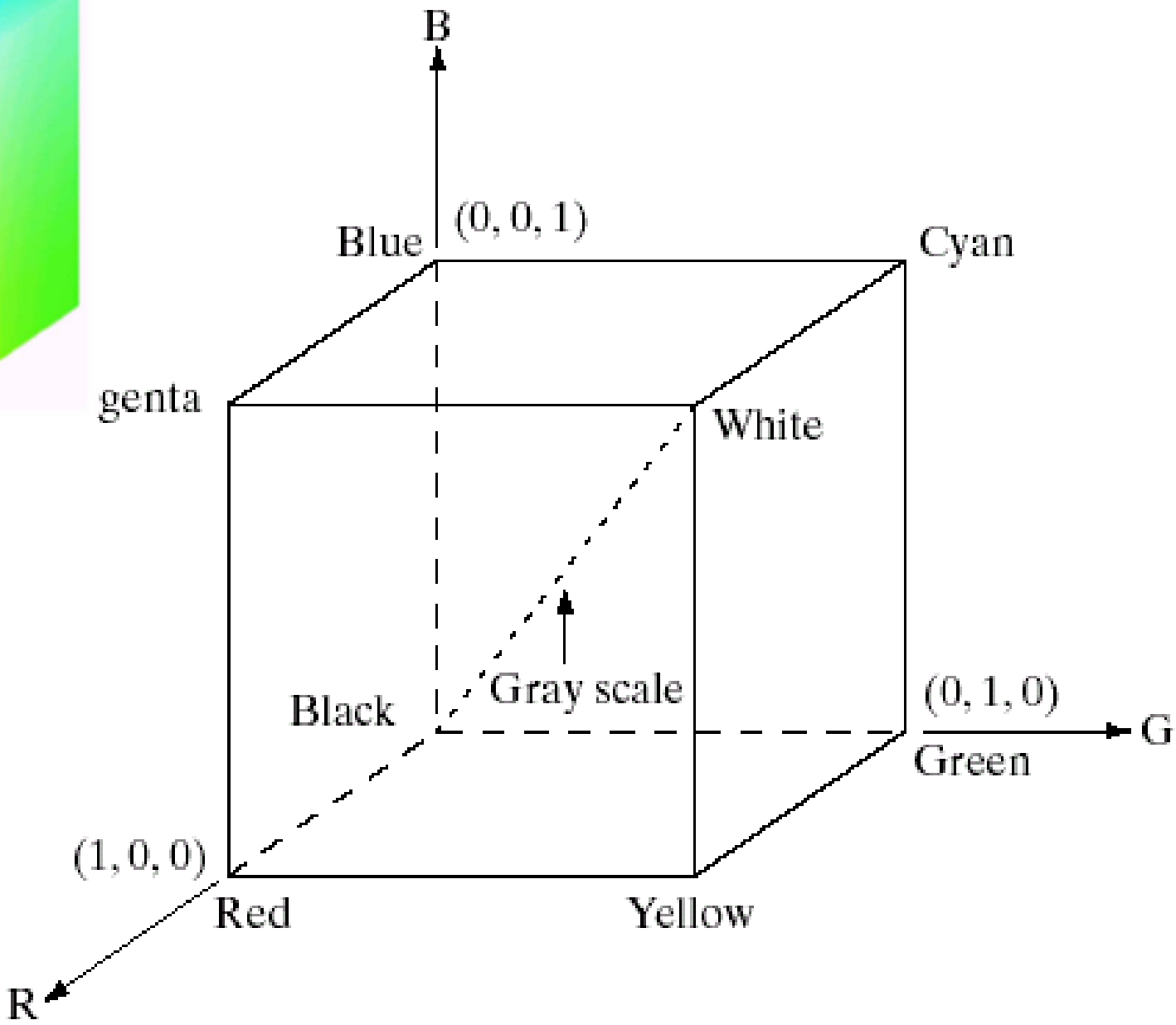
In the RGB model each colour appears in its primary spectral components of red, green and blue

The model is based on a Cartesian coordinate system

- RGB values are at 3 corners
- Cyan magenta and yellow are at three other corners
- Black is at the origin
- White is the corner furthest from the origin
- Different colours are points on or inside the cube represented by RGB vectors

RGB (cont…)





Images taken from Gonzalez & Woods, Digital Image Processing (2002)

RGB (cont…)

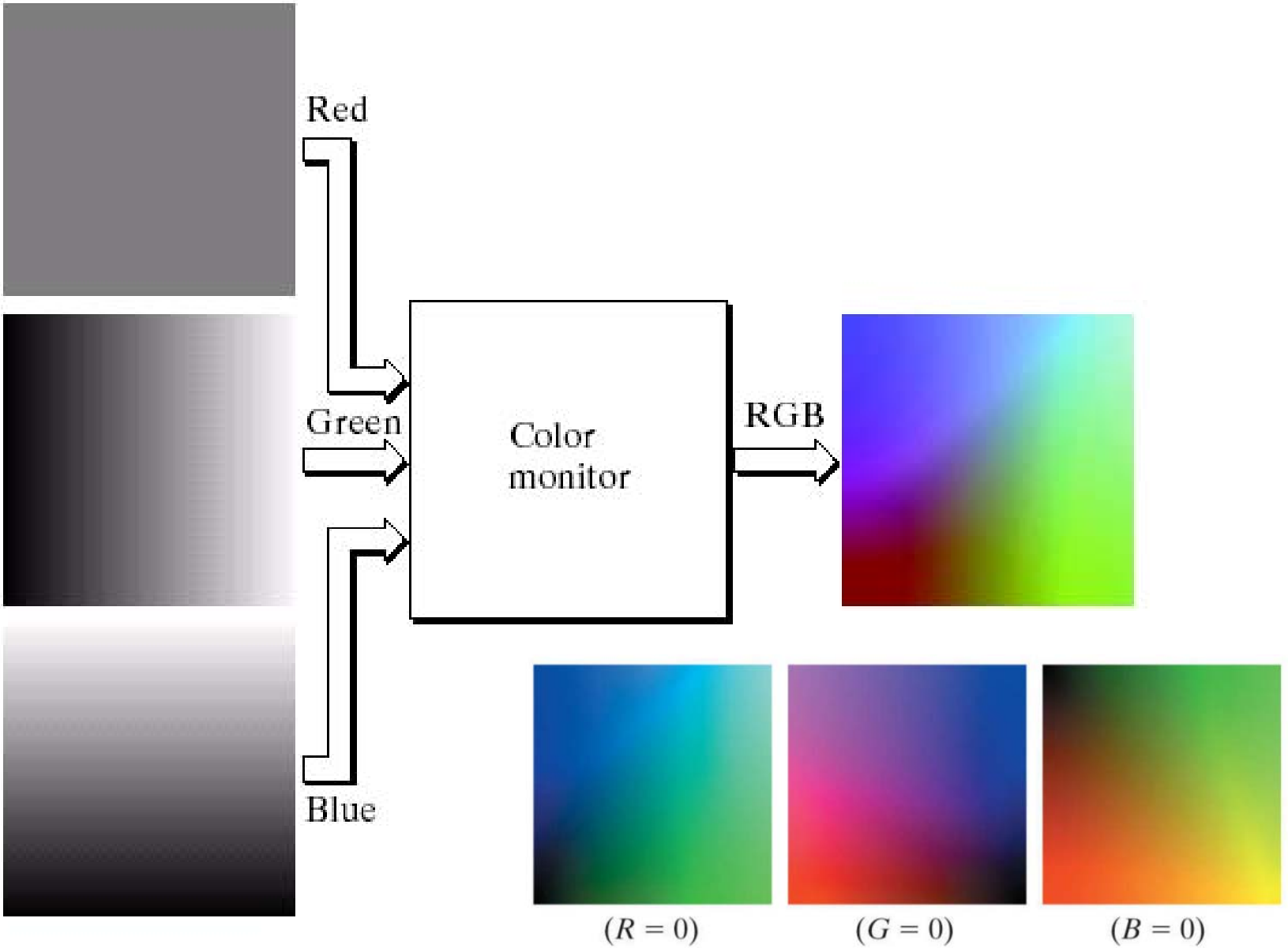
Images represented in the RGB colour model consist of three component images – one for each primary colour

When fed into a monitor these images are combined to create a composite colour image

The number of bits used to represent each pixel is referred to as the colour depth

A 24-bit image is often referred to as a full-colour image as it allows $\left(2^8\right)^3 = 16{,}777{,}216$ colours

RGB (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

The HSI Colour Model

RGB is useful for hardware implementations and is serendipitously related to the way in which the human visual system works

However, RGB is not a particularly intuitive way in which to describe colours

Rather when people describe colours they tend to use **hue**, **saturation** and **brightness**

RGB is great for colour generation, but HSI is great for colour description

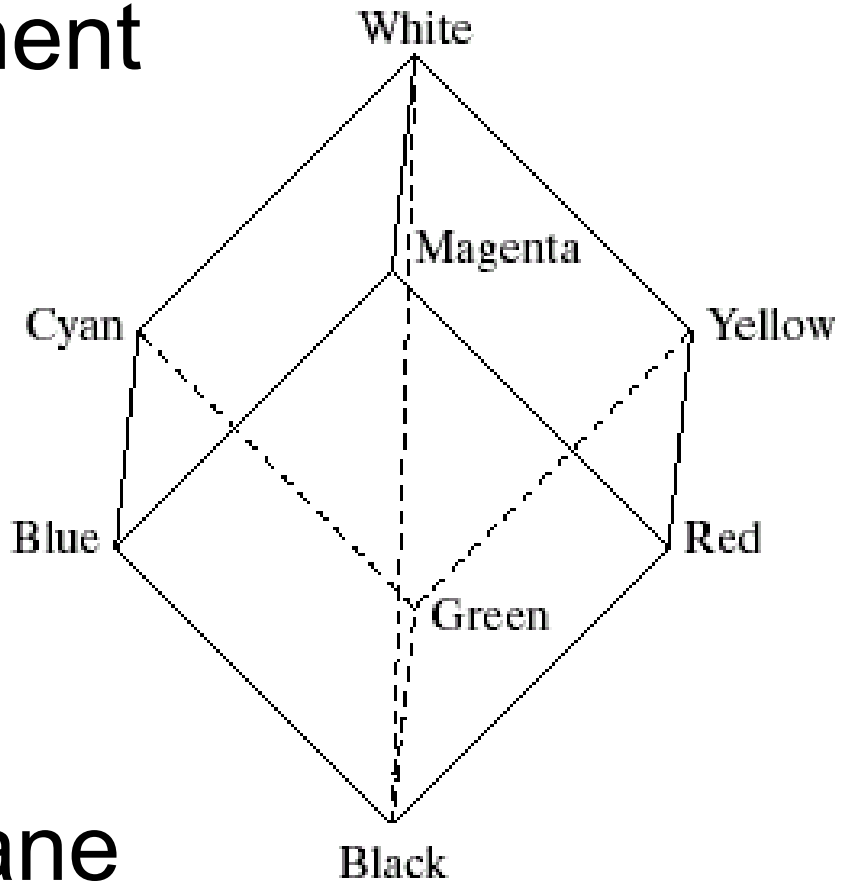
The HSI Colour Model (cont…)

The HSI model uses three measures to describe colours:

- **Hue:** A colour attribute that describes a pure colour (pure yellow, orange or red)
- **Saturation:** Gives a measure of how much a pure colour is diluted with white light
- **Intensity:** Brightness is nearly impossible to measure because it is so subjective. Instead we use intensity. Intensity is the same achromatic notion that we have seen in grey level images

HSI, Intensity & RGB (cont…)

Now the intensity component of any colour can be determined by passing a plane *perpendicular* to the intenisty axis and containing the colour point



The intersection of the plane with the intensity axis gives us the intensity component of the colour

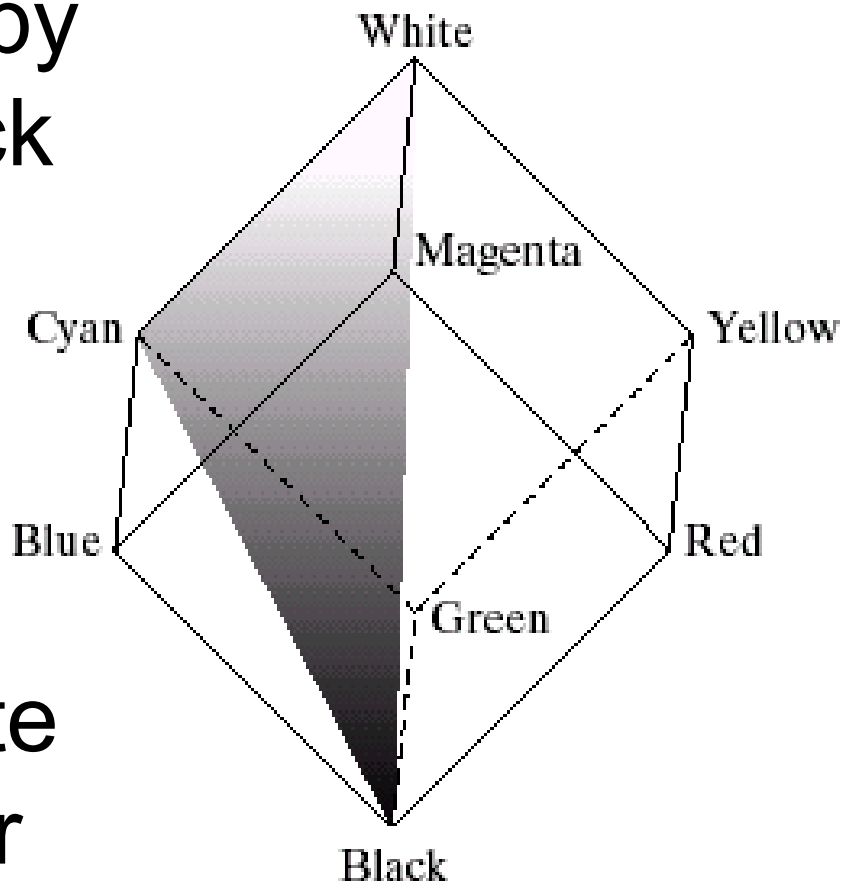
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

HSI, Hue & RGB

In a similar way we can extract the hue from the RGB colour cube

Consider a plane defined by the three points cyan, black and white

All points contained in this plane must have the same hue (cyan) as black and white cannot contribute hue information to a colour

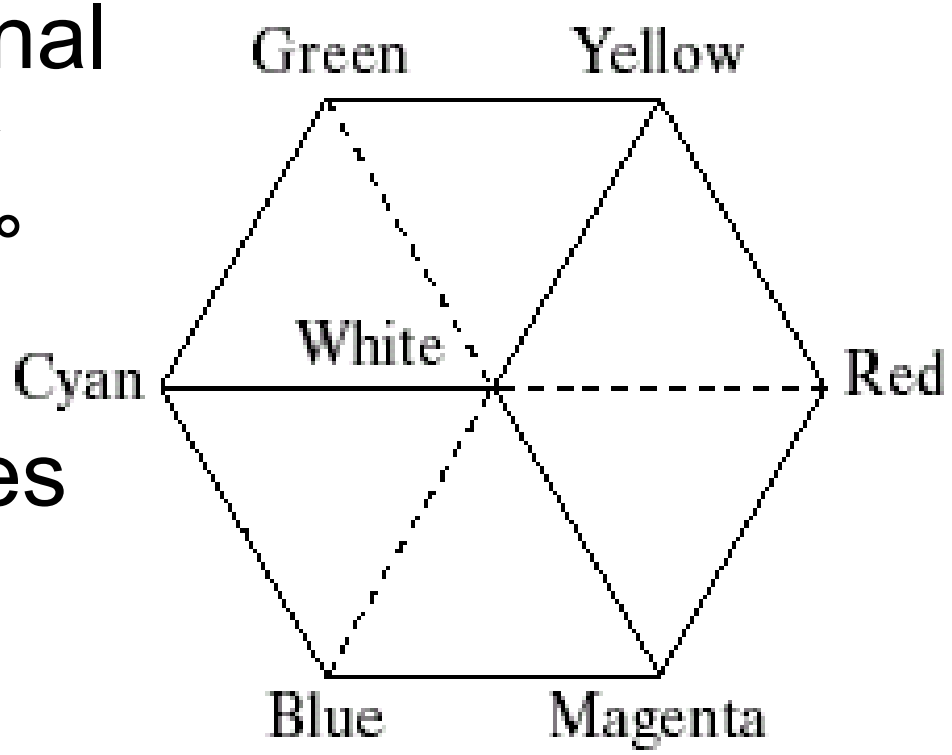


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

The HSI Colour Model

Consider if we look straight down at the RGB cube as it was arranged previously

We would see a hexagonal shape with each primary colour separated by 120° and secondary colours at 60° from the primaries

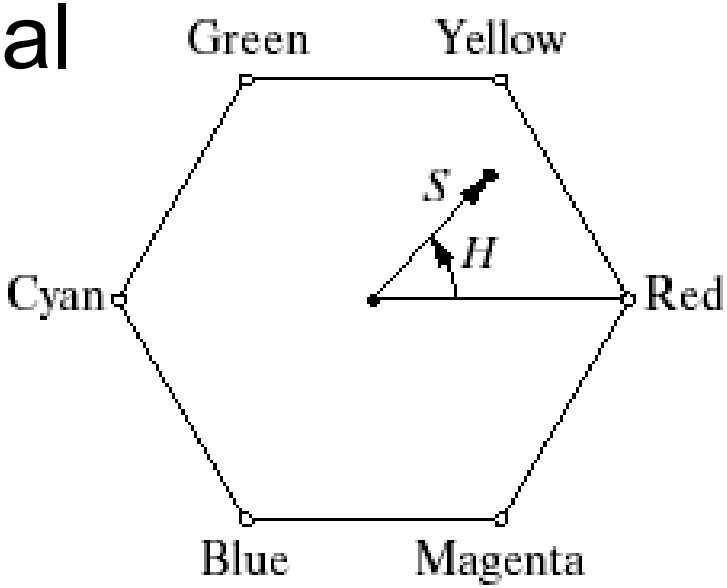


So the HSI model is composed of a vertical intensity axis and the locus of colour points that lie on planes perpendicular to that axis

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

The HSI Colour Model (cont…)

To the right we see a hexagonal shape and an arbitrary colour point

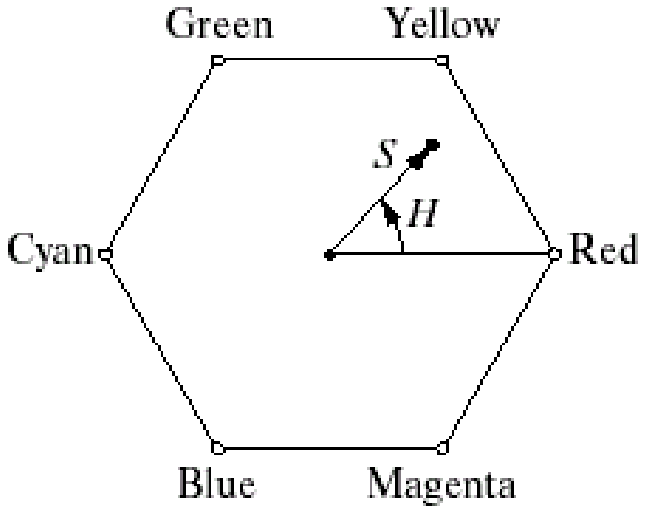


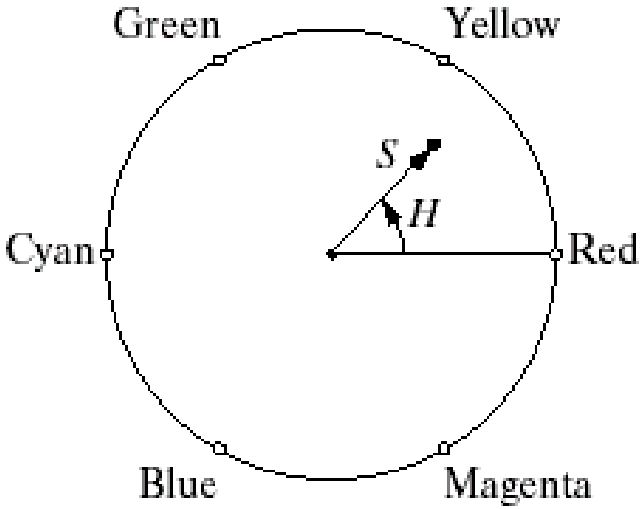
- The hue is determined by an angle from a reference point, usually red
- The saturation is the distance from the origin to the point
- The intensity is determined by how far up the vertical intenisty axis this hexagonal plane sits (not apparent from this diagram

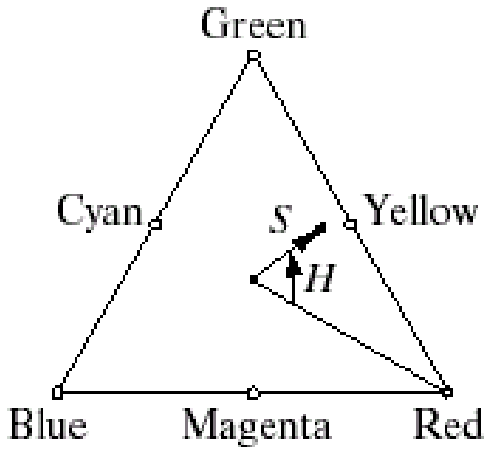
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

The HSI Colour Model (cont…)

Because the only important things are the angle and the length of the saturation vector this plane is also often represented as a circle or a triangle

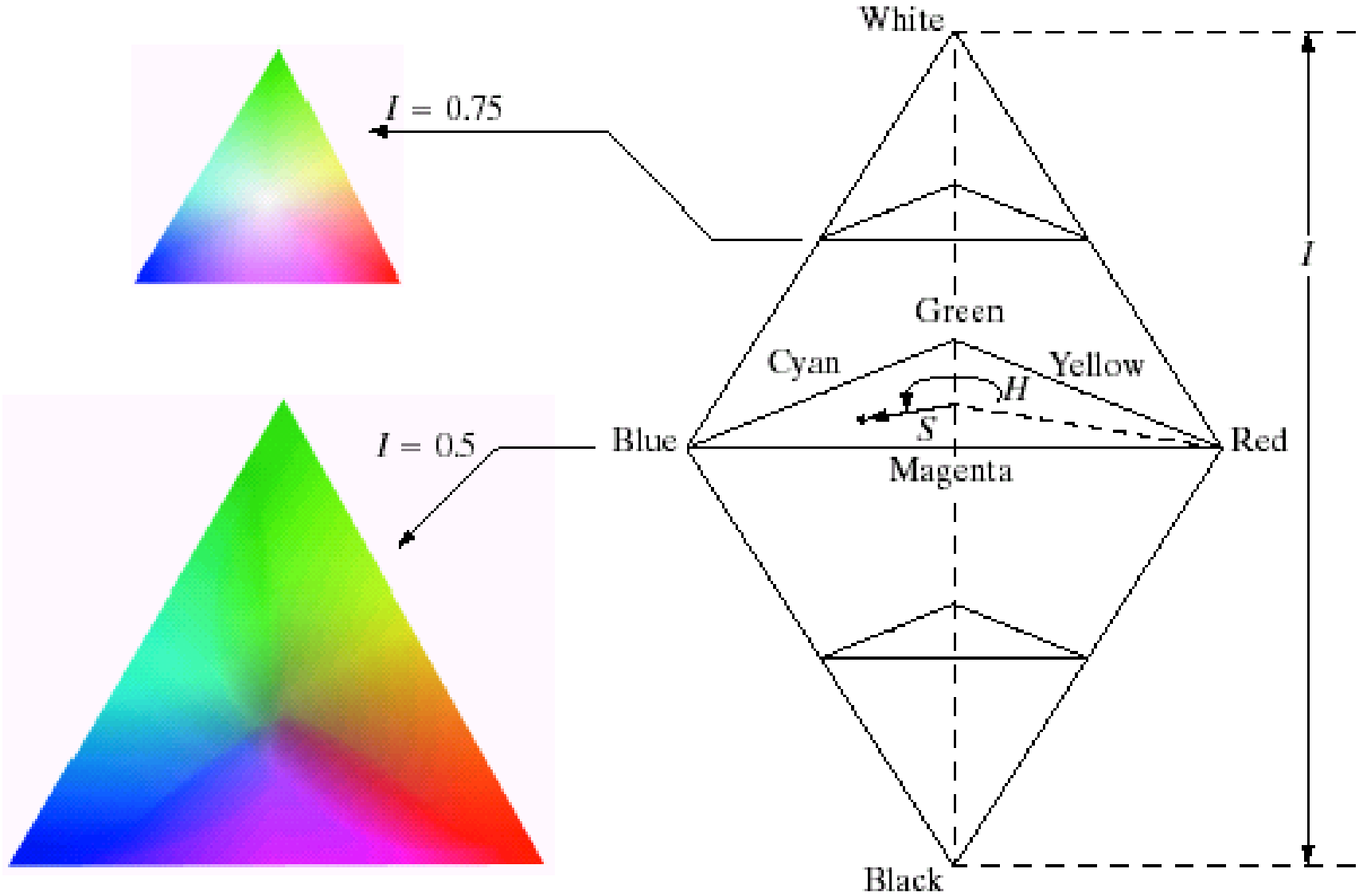






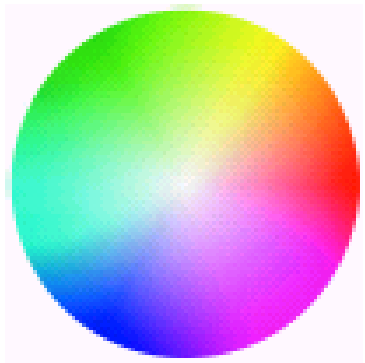
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

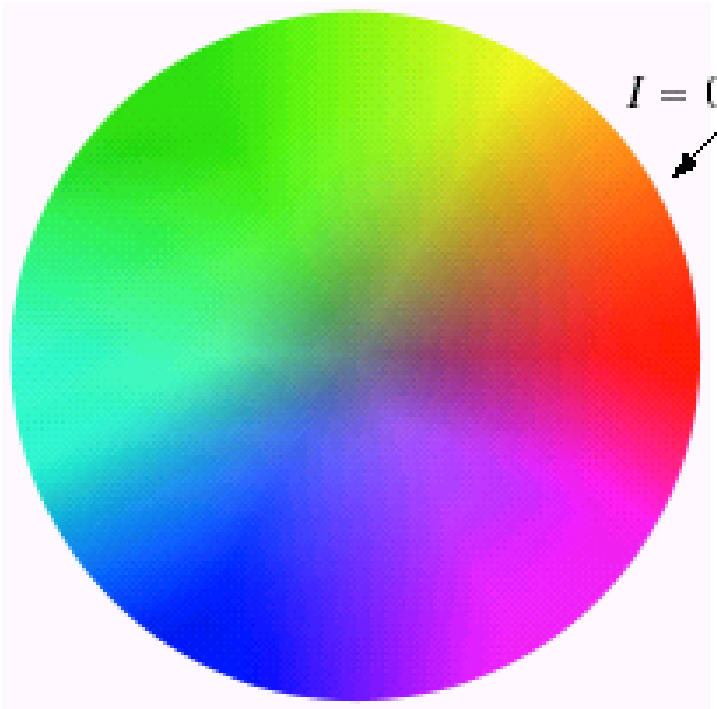
HSI Model Examples

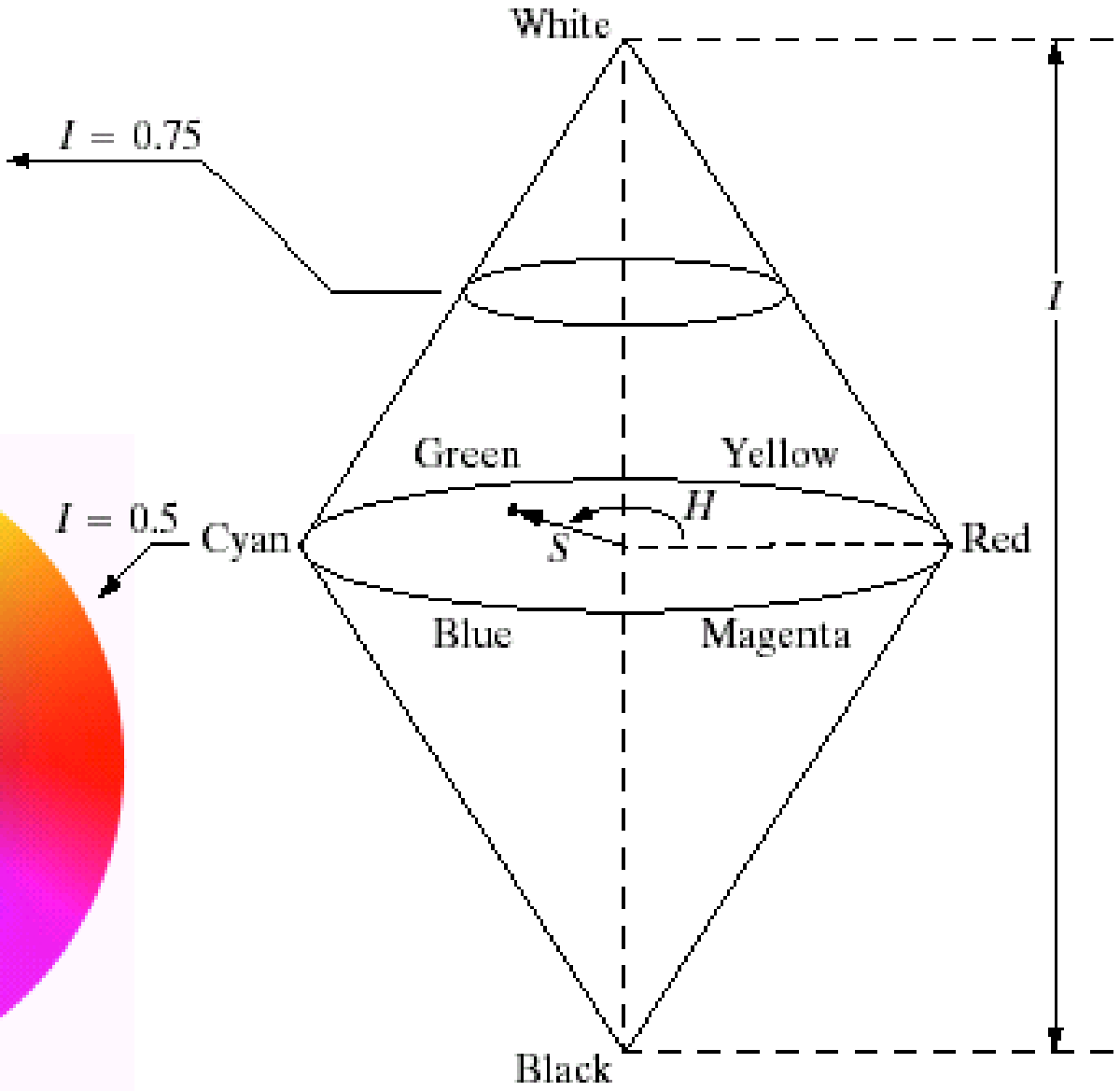


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

HSI Model Examples







Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Converting From RGB To HSI

Given a colour as R, G, and B its H, S, and I values are calculated as follows:

$$H = \begin{cases} \theta & if\ B \le G \\ 360 - \theta & if\ B > G \end{cases} \qquad \theta = \cos^{-1}\left\{ \frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{\frac{1}{2}}} \right\}$$

$$S = 1 - \frac{3}{(R+G+B)}\left[\min(R,G,B)\right] \qquad I = \frac{1}{3}(R+G+B)$$

Converting From HSI To RGB

Given a colour as H, S, and I it's R, G, and B values are calculated as follows:

- RG sector ($0 <= H < 120^\circ$)

$$R = I\left[1 + \frac{S\cos H}{\cos(60 - H)}\right] \qquad G = 3I - (R + B) \qquad B = I(1 - S)$$

- GB sector ($120^\circ <= H < 240^\circ$)

$$R = I(1 - S) \qquad G = I\left[1 + \frac{S\cos(H - 120)}{\cos(H - 60)}\right] \qquad B = 3I - (R + G)$$

Converting From HSI To RGB (cont…)

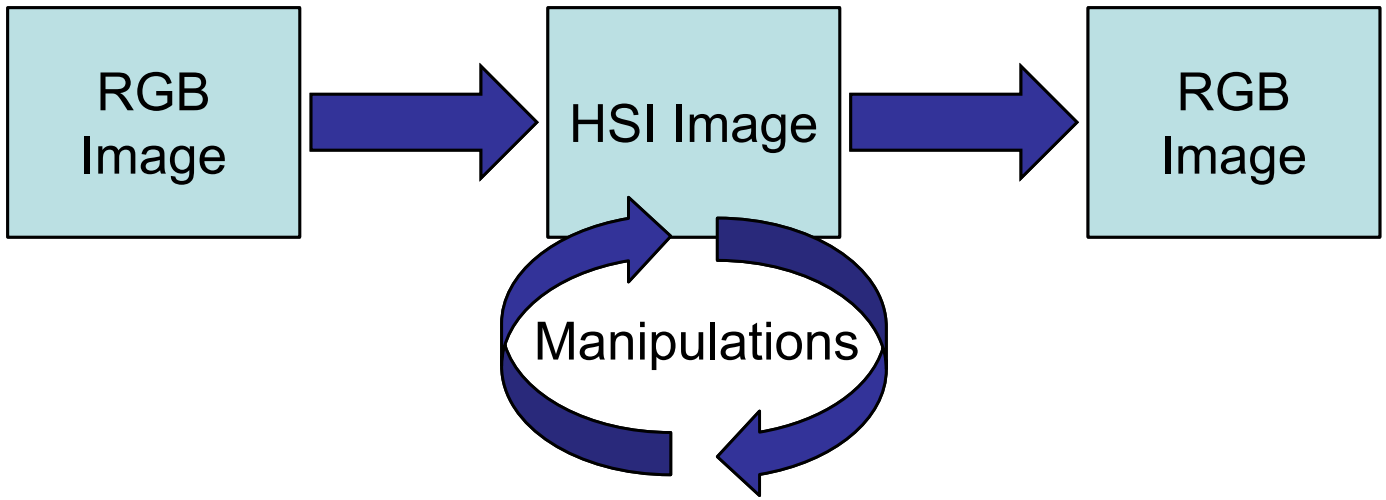
- BR sector ($240^\circ <= H <= 360^\circ$)

$$R = 3I - (G + B) \quad G = I(1 - S) \quad B = I\left[1 + \frac{S\cos(H - 240)}{\cos(H - 180)}\right]$$

Manipulating Images In The HSI Model

In order to manipulate an image under the HIS model we:

- First convert it from RGB to HIS
- Perform our manipulations under HSI
- Finally convert the image back from HSI to RGB

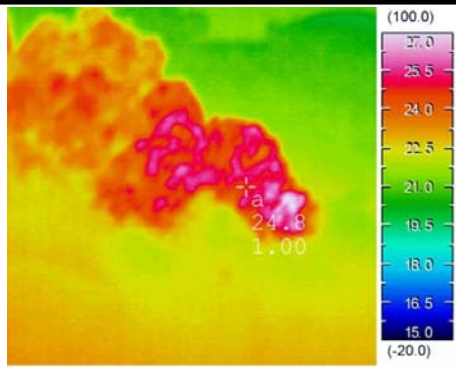


Pseudocolour Image Processing

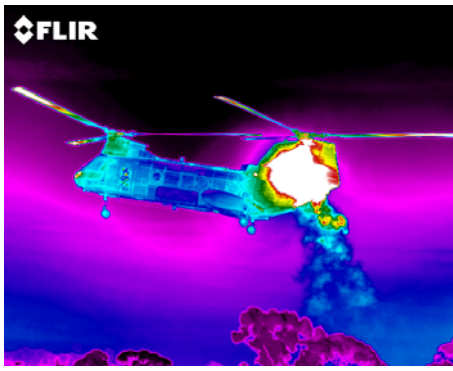
Pseudocolour (also called false colour) image processing consists of assigning colours to grey values based on a specific criterion

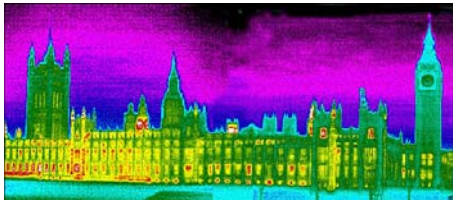
The principle use of pseudocolour image processing is for human visualisation

- Humans can discern between thousands of colour shades and intensities, compared to only about two dozen or so shades of grey









Pseudo Colour Image Processing – Intensity Slicing

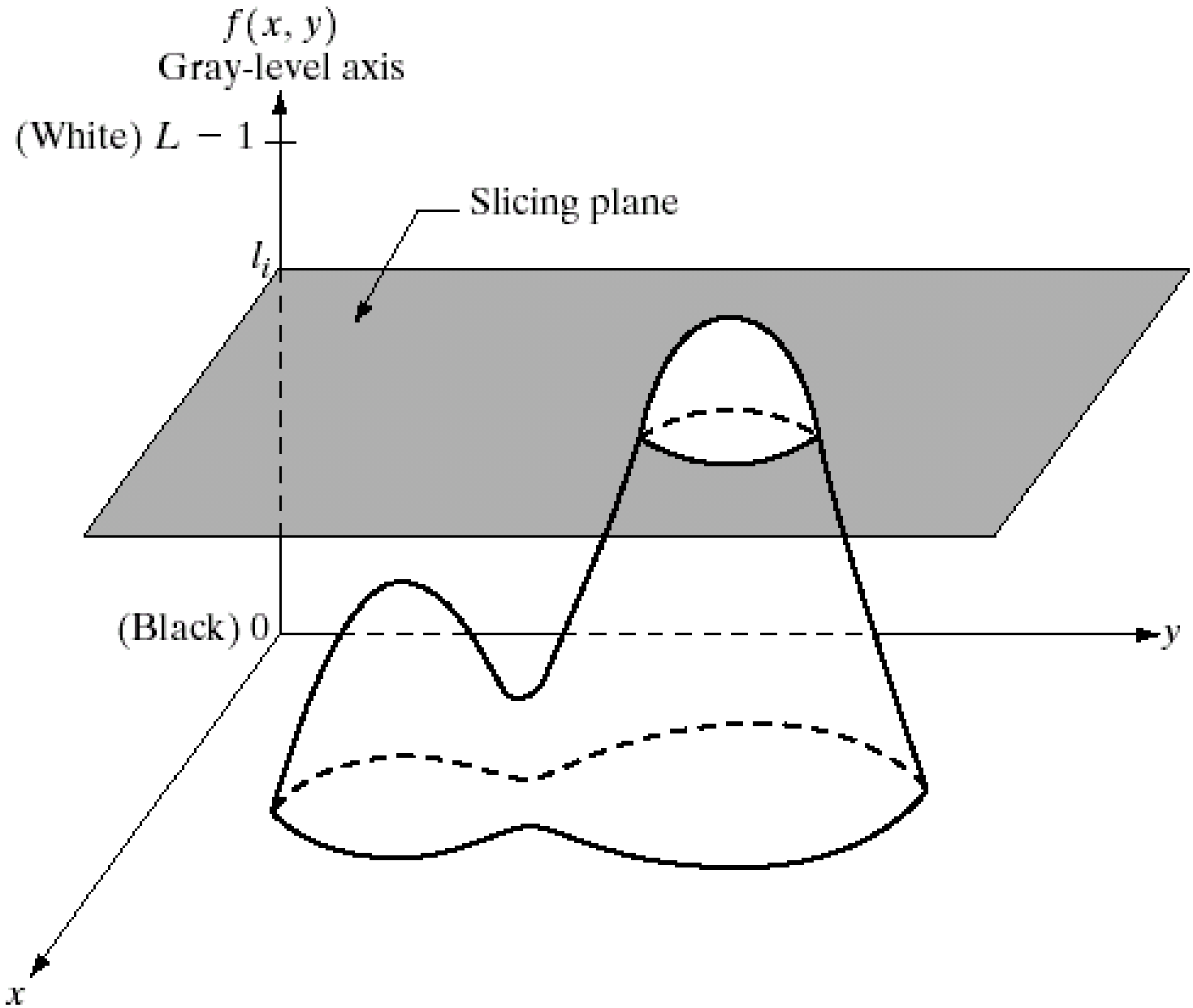
Intensity slicing and colour coding is one of the simplest kinds of pseudocolour image processing

First we consider an image as a 3D function mapping spatial coordinates to intensities (that we can consider heights)

Now consider placing planes at certain levels parallel to the coordinate plane

If a value is one side of such a plane it is rendered in one colour, and a different colour if on the other side

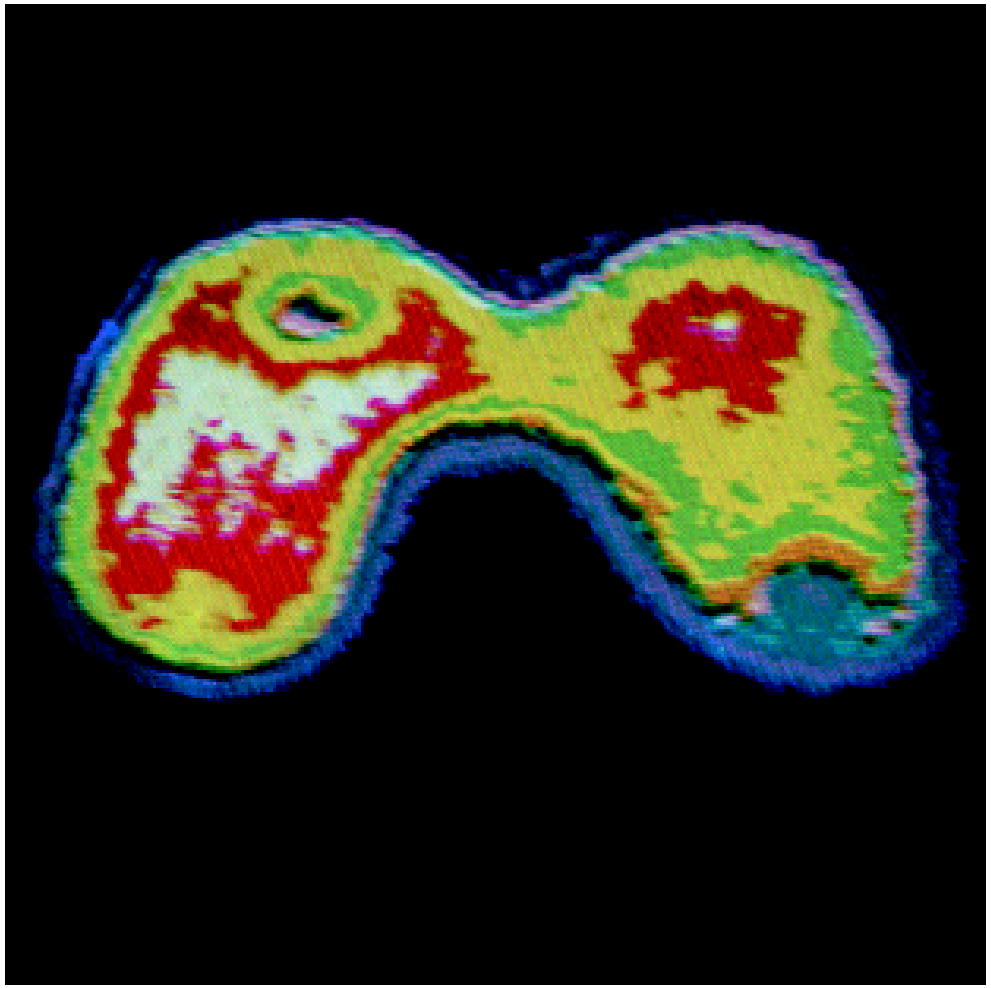
Pseudocolour Image Processing – Intensity Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

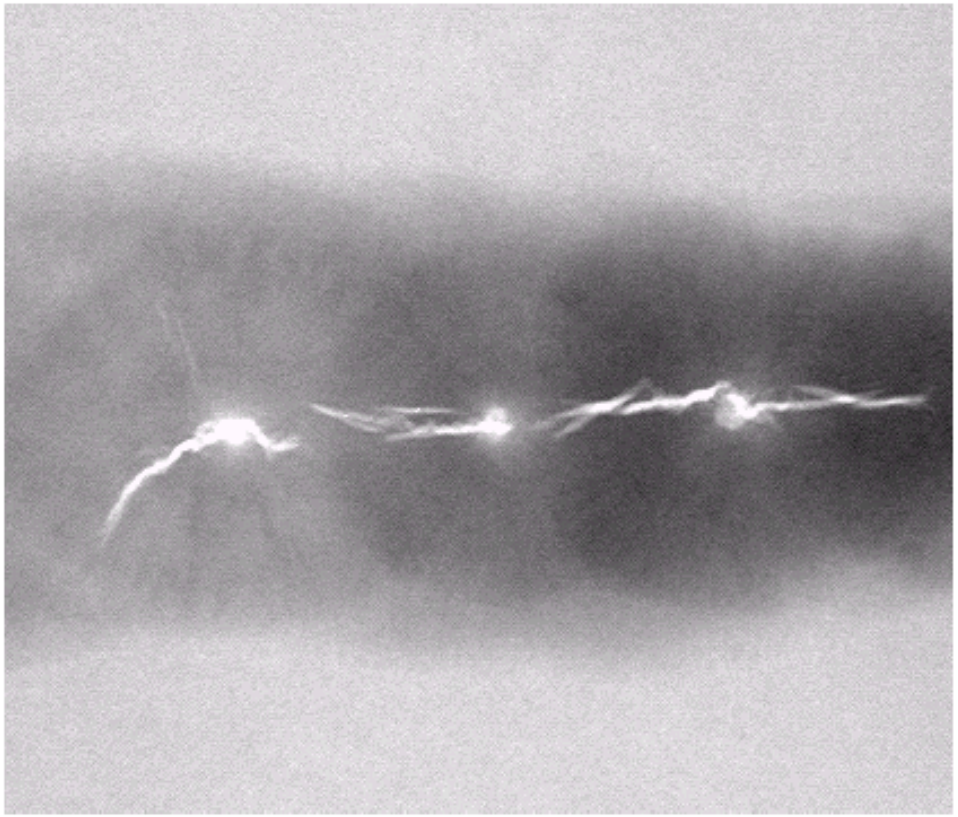
RGB -> HSI -> RGB (cont…)

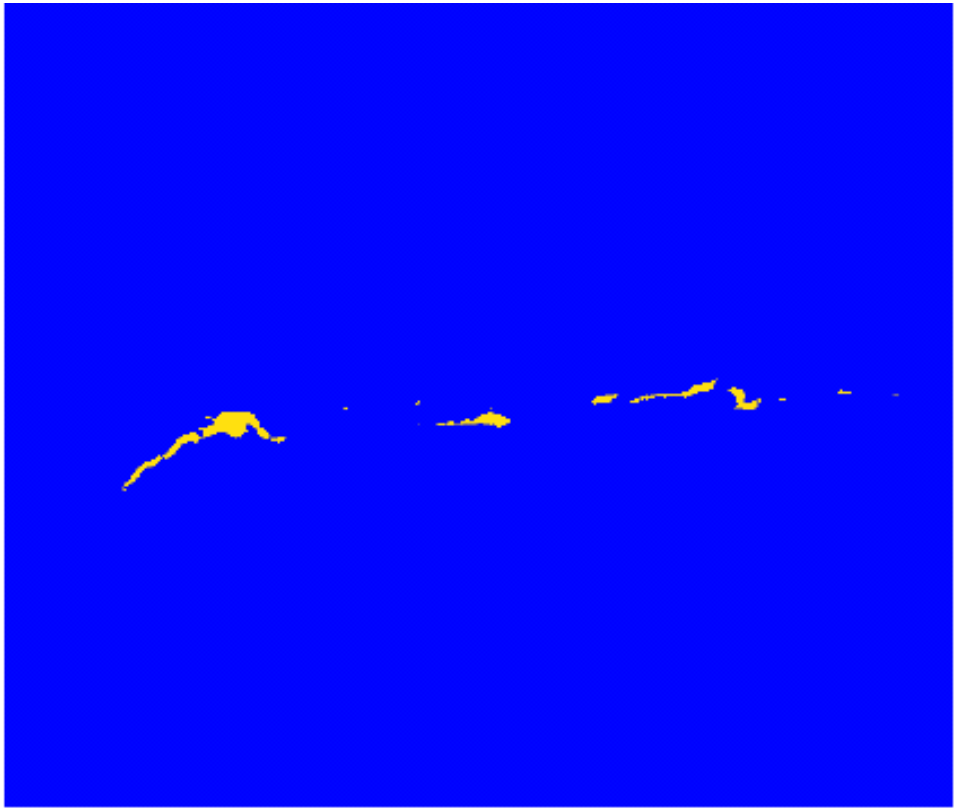




Images taken from Gonzalez & Woods, Digital Image Processing (2002)

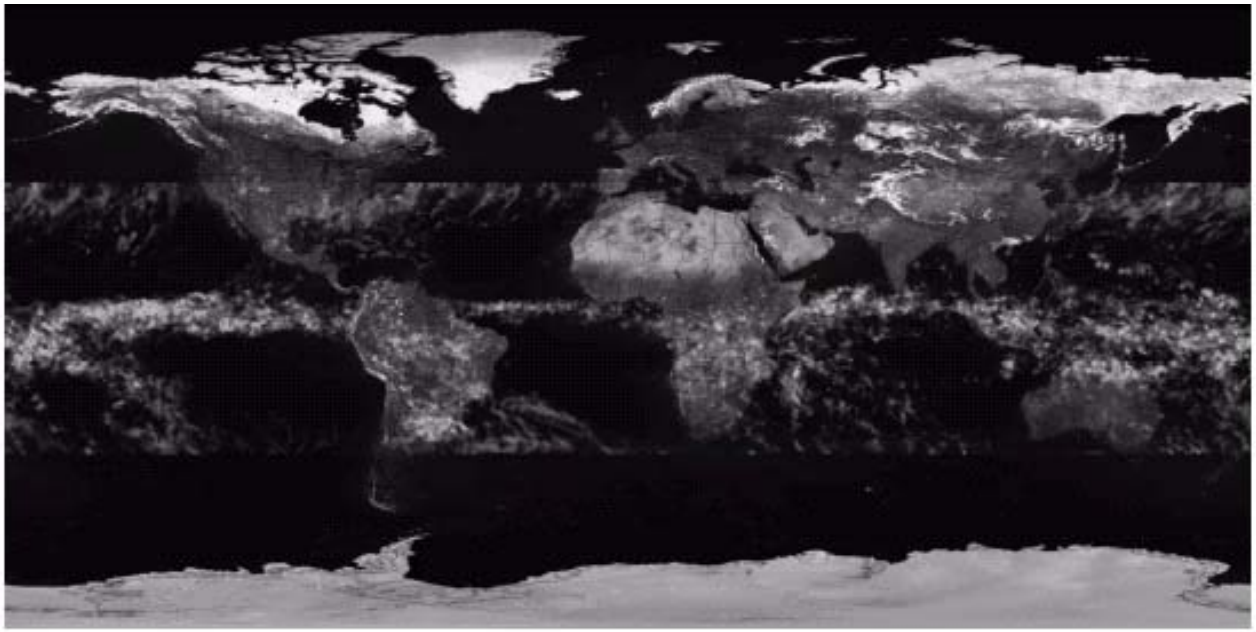
RGB -> HSI -> RGB (cont…)

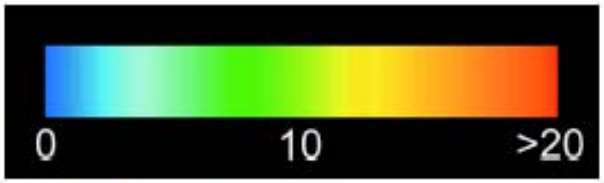


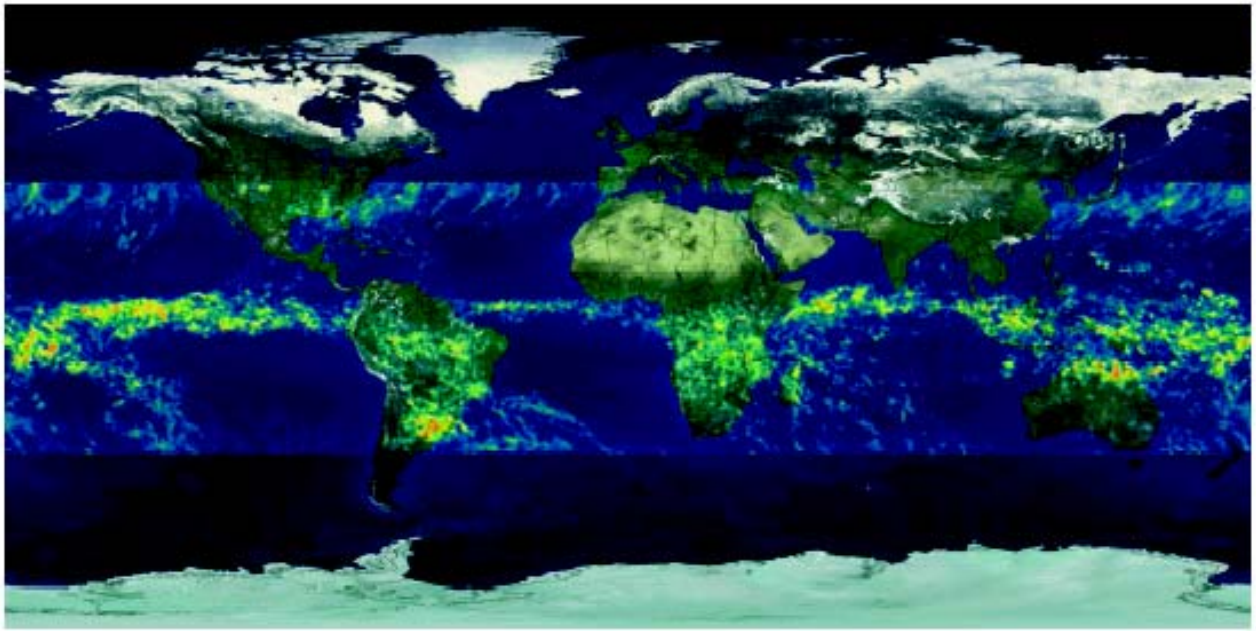


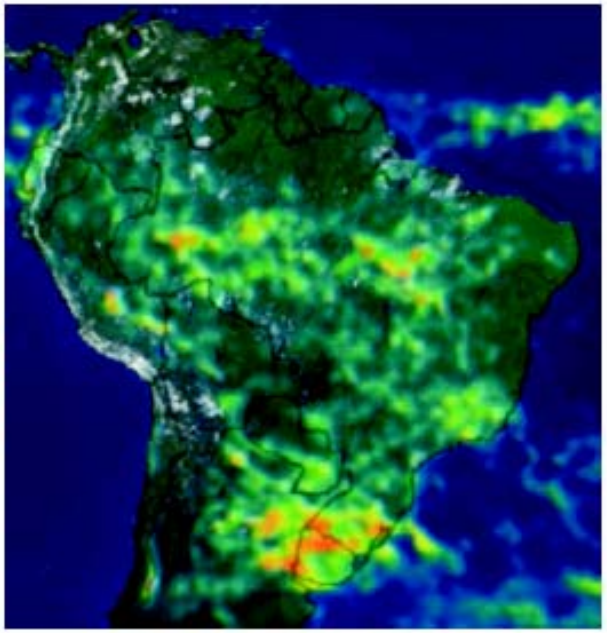
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

RGB -> HSI -> RGB (cont…)









Images taken from Gonzalez & Woods, Digital Image Processing (2002)

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Color Image Processing

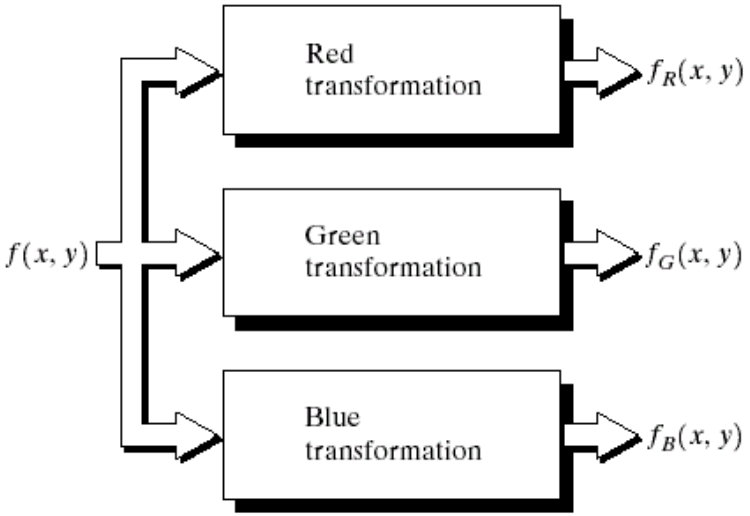
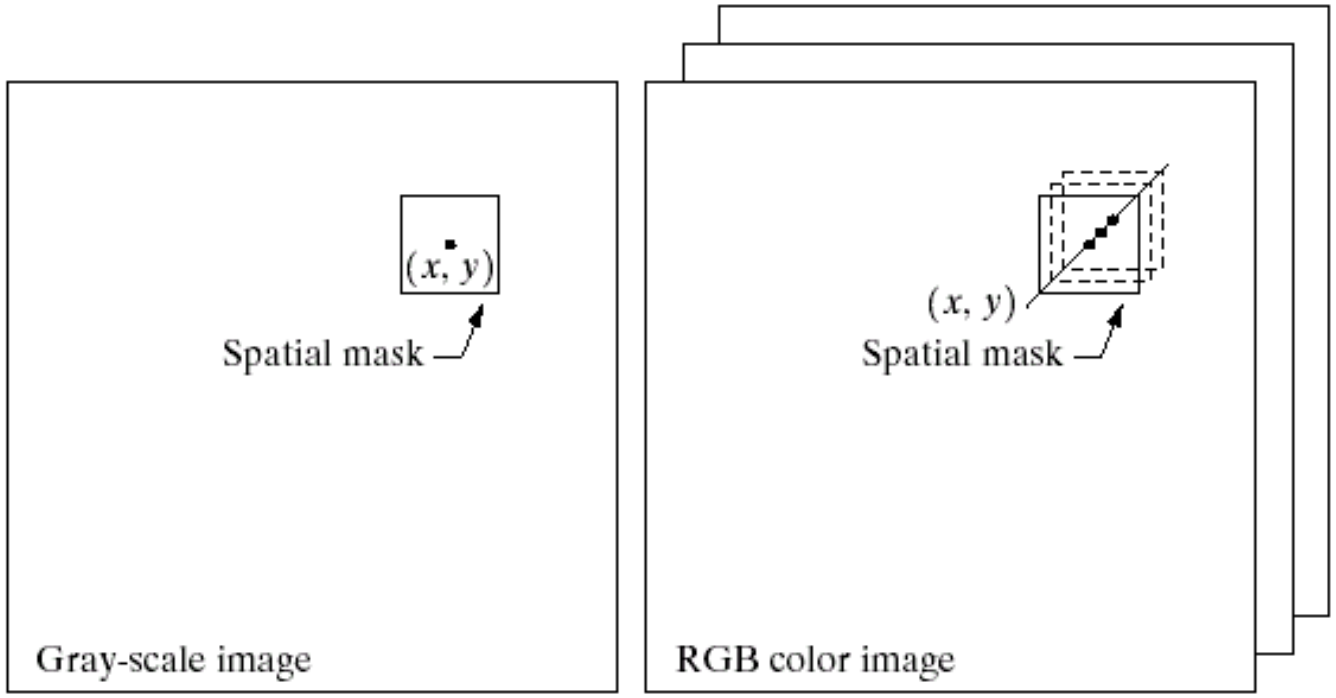


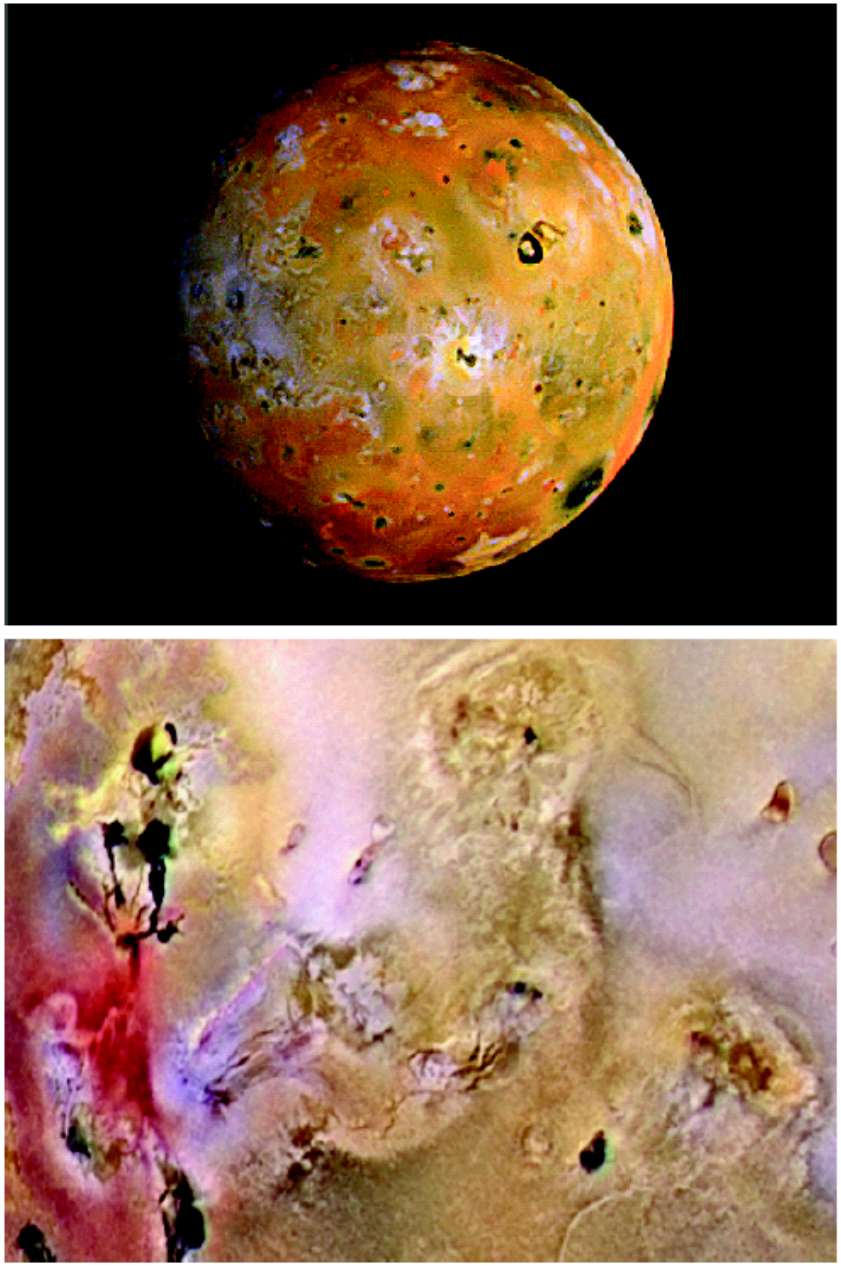
FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

a b

FIGURE 6.29
Spatial masks for gray-scale and RGB color images.



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a
b

FIGURE 6.28
(a) Pseudocolor rendition of Jupiter Moon Io. (b) A close-up. (Courtesy of NASA.)

Chapter 6
Color Image Processing



a b
c d

FIGURE 6.38
(a) RGB image.
(b) Red component image.
(c) Green component.
(d) Blue component.

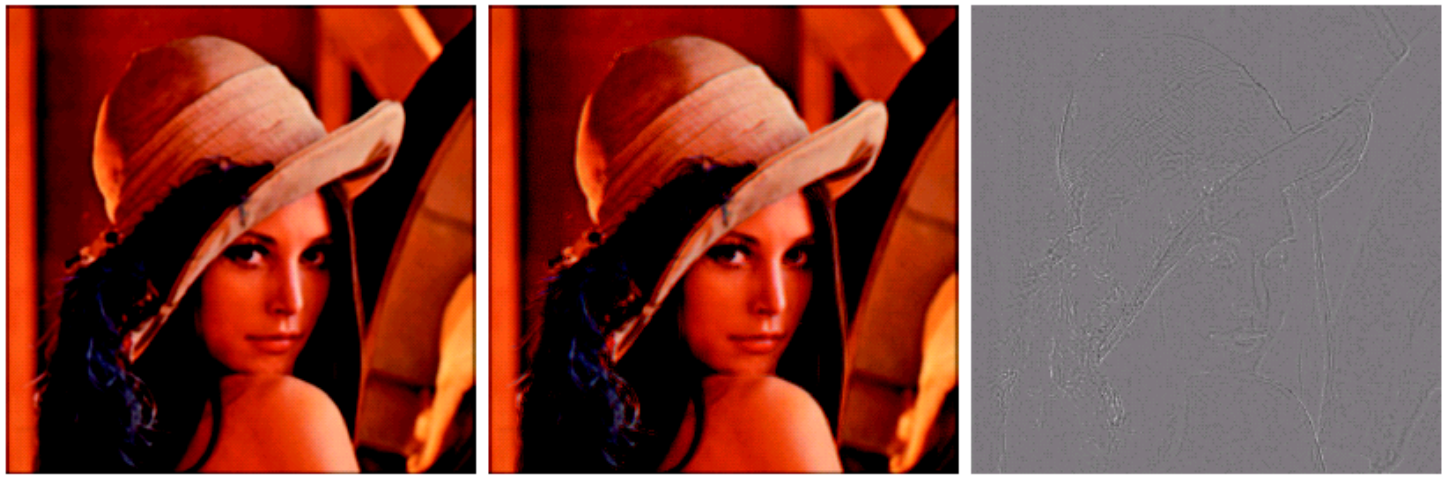
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a b c

FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

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a b c

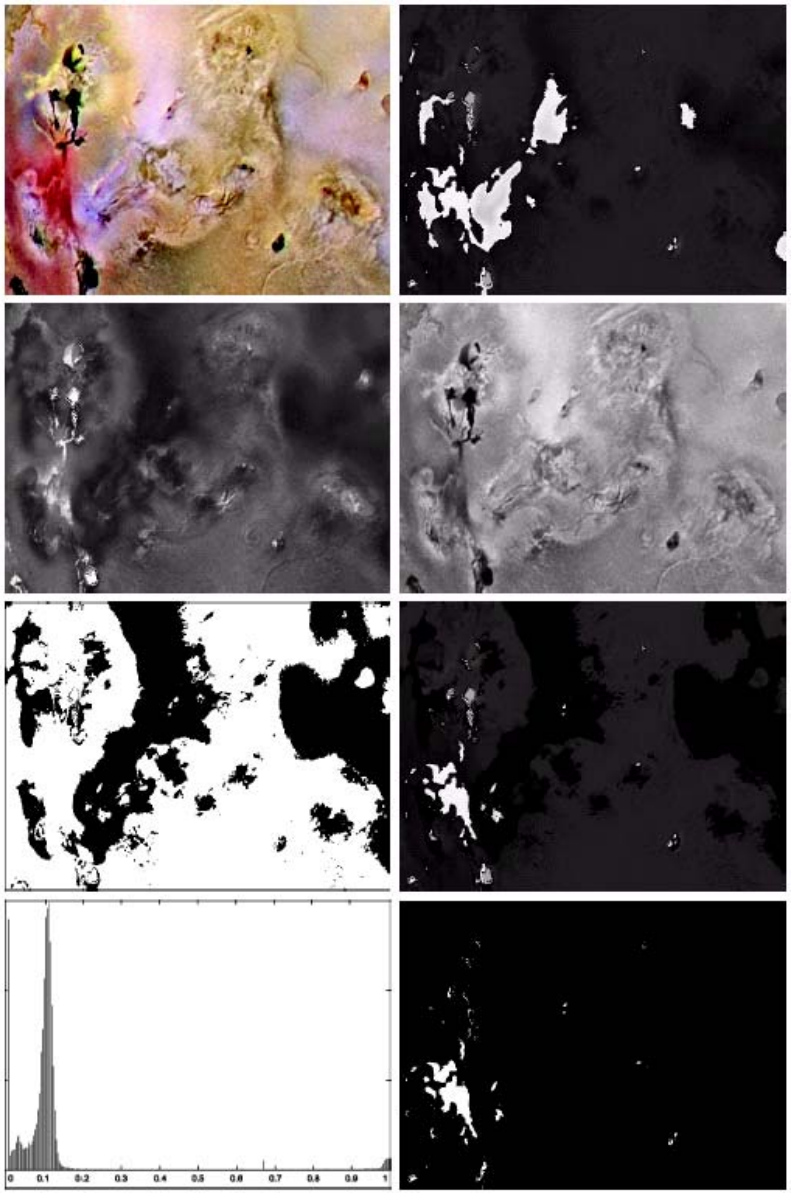
FIGURE 6.40 Image smoothing with a 5 × 5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.



a b c

FIGURE 6.39 Image sharpening using the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

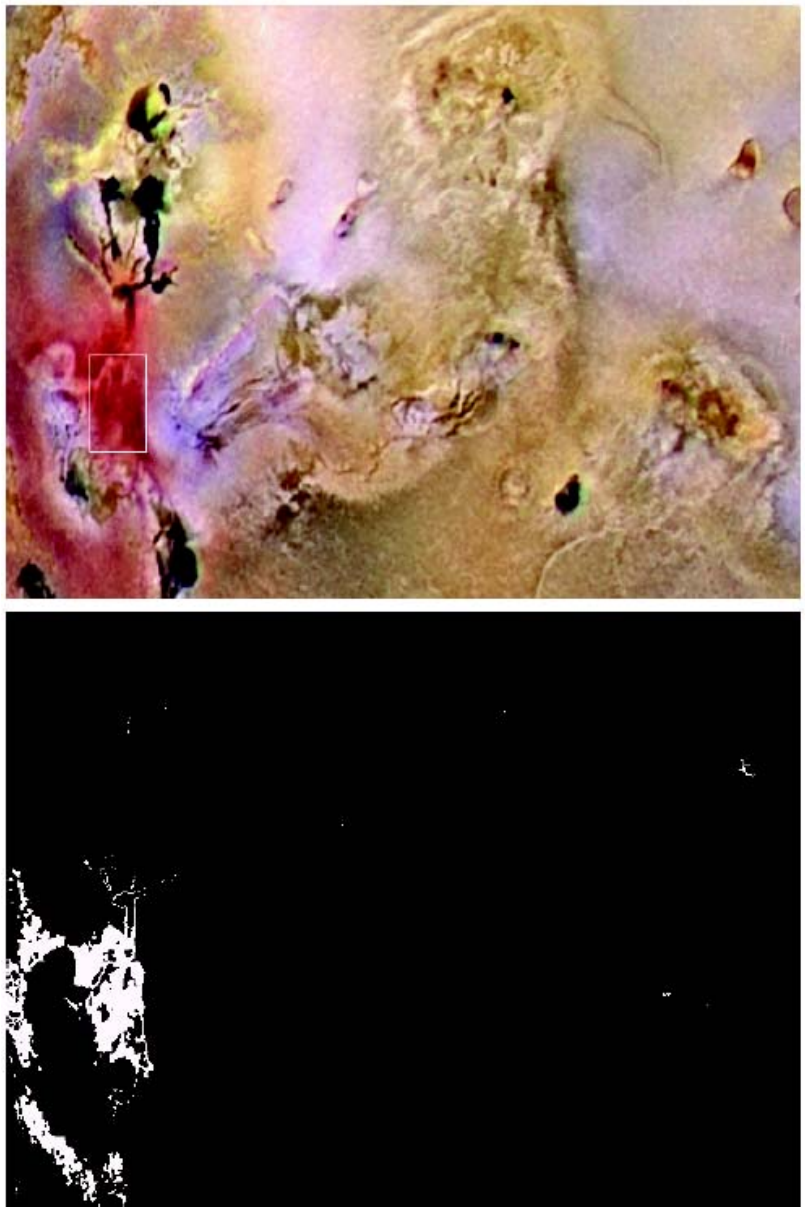
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a b
c d
e f
g h

FIGURE 6.42 Image segmentation in HSI space. (a) Original. (b) Hue. (c) Saturation. (d) Intensity. (e) Binary saturation mask (black = 0). (f) Product of (b) and (e). (g) Histogram of (f). (h) Segmentation of red components in (a).

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a
b

FIGURE 6.44
Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 6.42(h).

a b
c d

FIGURE 6.46
(a) RGB image.
(b) Gradient computed in RGB color vector space.
(c) Gradients computed on a per-image basis and then added.
(d) Difference between (b) and (c).





a b c

FIGURE 6.47 Component gradient images of the color image in Fig. 6.46. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.46(c).



a b
c d

FIGURE 6.48
(a)–(c) Red, green, and blue component images corrupted by additive Gaussian noise of mean 0 and variance 800. (d) Resulting RGB image. [Compare (d) with Fig. 6.46(a).]



a b c

FIGURE 6.49 HSI components of the noisy color image in Fig. 6.48(d). (a) Hue. (b) Saturation. (c) Intensity.



a b
c d

FIGURE 6.50
(a) RGB image with green plane corrupted by salt-and-pepper noise. (b) Hue component of HSI image. (c) Saturation component. (d) Intensity component.